Chapter 13

Clustering methods

13.1 Clustering

Clustering or *cluster analysis* is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar (in some sense) to each other than to those in other groups (clusters).

Clustering is a main task of exploratory data mining and used in many fields, including machine learning, pattern recognition, image analysis, information retrieval, bioinformatics, data compression, and computer graphics. It can be achieved by various algorithms that differ significantly in their notion of what constitutes a cluster and how to efficiently find them. Popular notions of clusters include groups with small distances between cluster members, dense areas of the data space, etc.

13.1.1 Examples of data with natural clusters

In many applications, there will naturally be several groups or clusters in samples.

1. Consider the case of optical character recognition: There are two ways of writing the digit 7; the American writing is '7', whereas the European writing style has a horizontal bar in the middle (something like 7̶). In such a case, when the sample contains examples from both continents, the sample will contain two clusters or groups one corresponding to the American 7 and the other corresponding to the European 7̶.

2. In speech recognition, where the same word can be uttered in different ways, due to different pronunciation, accent, gender, age, and so forth, there is not a single, universal prototype. In a large sample of utterances of a specific word, All the different ways should be represented in the sample.

13.2 k-means clustering

13.2.1 Outline

The k-means clustering algorithm is one of the simplest unsupervised learning algorithms for solving the clustering problem.

Let it be required to classify a given data set into a certain number of clusters, say, k clusters. We start by choosing k points arbitrarily as the "centres" of the clusters, one for each cluster. We then associate each of the given data points with the nearest centre. We now take the averages of the data points associated with a centre and replace the centre with the average, and this is done for each of the centres. We repeat the process until the centres converge to some fixed points. The data points nearest to the centres form the various clusters in the dataset. Each cluster is represented by the associated centre.

13.2.2 Example

We illustrate the algorithm in the case where there are only two variables so that the data points and cluster centres can be geometrically represented by points in a coordinate plane. The distance between the points (x_1, x_2) and (y_1, y_2) will be calculated using the familiar distance formula of elementary analytical geometry:

$$\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

Problem

Use k-means clustering algorithm to divide the following data into two clusters and also compute the the representative data points for the clusters.

x_1	1	2	2	3	4	5
x_2	1	1	3	2	3	5

Table 13.1: Data for k-means algorithm example

Solution

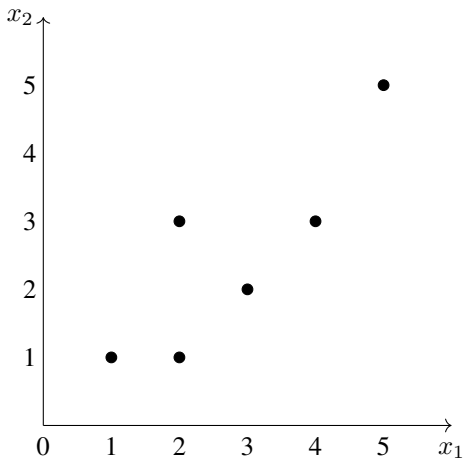


Figure 13.1: Scatter diagram of data in Table 13.1

1. In the problem, the required number of clusters is 2 and we take k = 2.
2. We choose two points arbitrarily as the initial cluster centres. Let us choose arbitrarily (see Figure 13.2)
$$\vec{v}_1 = (2, 1), \quad \vec{v}_2 = (2, 3).$$
3. We compute the distances of the given data points from the cluster centers.

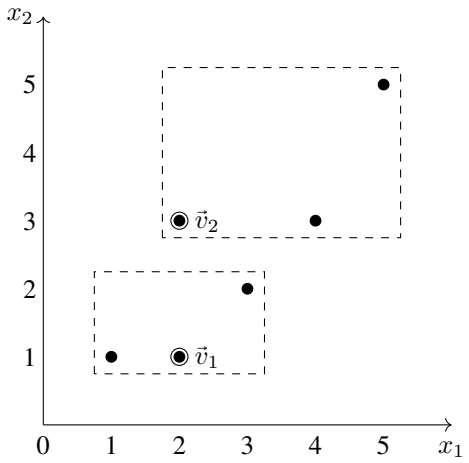


Figure 13.2: Initial choice of cluster centres and the resulting clusters

$\vec{x}_i$	Data point	Distance from $\vec{v}_1 = (2,1)$	Distance from $\vec{v}_2 = (2,3)$	Minimum distance	Assigned center
$\vec{x}_1$	$(1,1)$	1	2.24	1	$\vec{v}_1$
$\vec{x}_2$	$(2,1)$	0	2	0	$\vec{v}_1$
$\vec{x}_3$	$(2,3)$	2	0	0	$\vec{v}_2$
$\vec{x}_4$	$(3,2)$	1.41	1.41	0	$\vec{v}_1$
$\vec{x}_5$	$(4,3)$	2.82	2	2	$\vec{v}_2$
$\vec{x}_6$	$(5,5)$	5	3.61	3.61	$\vec{v}_2$

(The distances of $\vec{x}_4$ from $\vec{v}_1$ and $\vec{v}_2$ are equal. We have assigned $\vec{v}_1$ to $\vec{x}_4$ arbitrarily.)

This divides the data into two clusters as follows (see Figure 13.2):

Cluster 1: $\{\vec{x}_1, \vec{x}_2, \vec{x}_4\}$ represented by $\vec{v}_1$

Number of data points in Cluster 1: $c_1 = 3$.

Cluster 2 : $\{\vec{x}_3, \vec{x}_5, \vec{x}_6\}$ represented by $\vec{v}_2$

Number of data points in Cluster 2: $c_2 = 3$.

4. The cluster centres are recalculated as follows:

$$\begin{aligned}\vec{v}_1 &= \frac{1}{c_1}(\vec{x}_1 + \vec{x}_2 + \vec{x}_4) \\ &= \frac{1}{3}(\vec{x}_1 + \vec{x}_2 + \vec{x}_4) \\ &= (2.00, 1.33) \\ \vec{v}_2 &= \frac{1}{c_2}(\vec{x}_3 + \vec{x}_5 + \vec{x}_6) \\ &= \frac{1}{3}(\vec{x}_3 + \vec{x}_5 + \vec{x}_6) \\ &= (3.67, 3.67)\end{aligned}$$

5. We compute the distances of the given data points from the new cluster centers.

$\vec{x}_i$	Data point	Distance from $\vec{v}_1 = (2,1)$	Distance from $\vec{v}_2 = (2,3)$	Minimum distance	Assigned center
$\vec{x}_1$	$(1,1)$	1.05	3.77	1.05	$\vec{v}_1$
$\vec{x}_2$	$(2,1)$	0.33	3.14	0.33	$\vec{v}_1$
$\vec{x}_3$	$(2,3)$	1.67	1.80	1.67	$\vec{v}_1$
$\vec{x}_4$	$(3,2)$	1.20	1.80	1.20	$\vec{v}_1$
$\vec{x}_5$	$(4,3)$	2.60	0.75	0.75	$\vec{v}_2$
$\vec{x}_6$	$(5,5)$	4.74	1.89	1.89	$\vec{v}_2$

This divides the data into two clusters as follows (see Figure 13.4):

Cluster 1 : $\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}$ represented by $\vec{v}_1$

Number of data points in Cluster 1: $c_1 = 4$.

Cluster 2 : $\{\vec{x}_5, \vec{x}_6\}$ represented by $\vec{v}_2$

Number of data points in Cluster 1: $c_2 = 2$.

6. The cluster centres are recalculated as follows:

$$\begin{aligned}\vec{v}_1 &== \frac{1}{c_1}(\vec{x}_1 + \vec{x}_2 + \vec{x}_3 + \vec{x}_4) \\ &= \frac{1}{4}(\vec{x}_1 + \vec{x}_2 + \vec{x}_3 + \vec{x}_4) \\ &= (2.00, 1.33) \\ \vec{v}_2 &= \frac{1}{2}(\vec{x}_5 + \vec{x}_6) = (3.67, 3.67)\end{aligned}$$

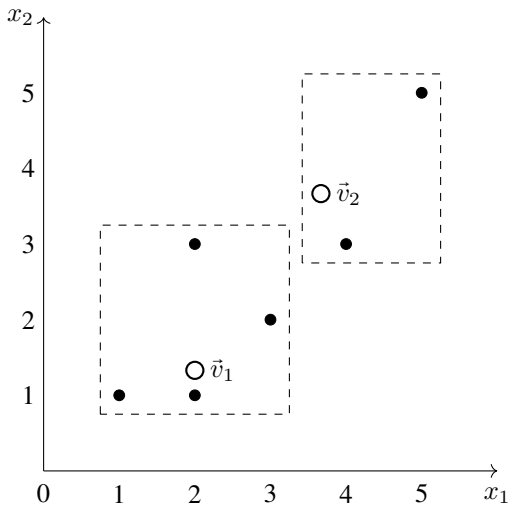


Figure 13.3: Cluster centres after first iteration and the corresponding clusters

7. We compute the distances of the given data points from the new cluster centers.
   4.609772 3.905125 2.692582 2.500000 1.118034 1.118034

$\vec{x}_i$	Data point	Distance from $\vec{v}_1 = (2,1)$	Distance from $\vec{v}_2 = (2,3)$	Minimum distance	Assigned center
$\vec{x}_1$	$(1,1)$	1.25	4.61	1.25	$\vec{v}_1$
$\vec{x}_2$	$(2,1)$	0.75	3.91	0.75	$\vec{v}_1$
$\vec{x}_3$	$(2,3)$	1.25	2.69	1.25	$\vec{v}_1$
$\vec{x}_4$	$(3,2)$	1.03	2.50	1.03	$\vec{v}_1$
$\vec{x}_5$	$(4,3)$	2.36	1.12	1.12	$\vec{v}_2$
$\vec{x}_6$	$(5,5)$	4.42	1.12	1.12	$\vec{v}_2$

This divides the data into two clusters as follows (see Figure **??**):

Cluster 1 : $\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}$ represented by $\vec{v}_1$

Number of data points in Cluster 1: $c_1 = 4$.

Cluster 2 : $\{\vec{x}_5, \vec{x}_6\}$ represented by $\vec{v}_2$

Number of data points in Cluster 1: $c_1 = 2$.

8. The cluster centres are recalculated as follows:

$$\begin{aligned}\vec{v}_1 &= \frac{1}{c_1}(\vec{x}_1 + \vec{x}_2 + \vec{x}_3 + \vec{x}_4) \\ &= \frac{1}{4}(\vec{x}_1 + \vec{x}_2 + \vec{x}_3 + \vec{x}_4) \\ &= (2.00, 1.75) \\ \vec{v}_2 &= \frac{1}{c_2}(\vec{x}_5 + \vec{x}_6) \\ &= \frac{1}{2}(\vec{x}_5 + \vec{x}_6) \\ &= (4.00, 4.50)\end{aligned}$$

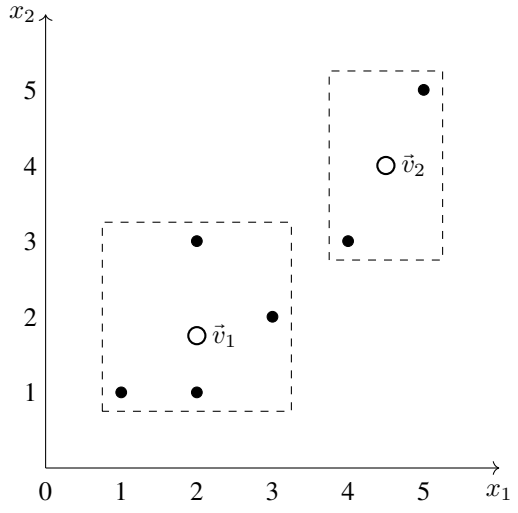


Figure 13.4: New cluster centres and the corresponding clusters

9. This divides the data into two clusters as follows (see Figure **??**):

Cluster 1 : $\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}$ represented by $\vec{v}_1$

Cluster 2 : $\{\vec{x}_5, \vec{x}_6\}$ represented by $\vec{v}_2$

10. The cluster centres are recalculated as follows:

$$\vec{v}_1 = \frac{1}{4}(\vec{x}_1 + \vec{x}_2 + \vec{x}_3 + \vec{x}_4) = (2.00, 1.75)$$
$$\vec{v}_2 = \frac{1}{2}(\vec{x}_5 + \vec{x}_6) = (4.00, 4.50)$$

    We note that these are identical to the cluster centres calculated in Step 8. So there will be no reassignment of data points to different clusters and hence the computations are stopped here.

11. Conclusion: The k means clustering algorithm with $k = 2$ applied to the dataset in Table 13.1 yields the following clusters and the associated cluster centres:

    Cluster 1 : $\{\vec{x}_1, \vec{x}_2, \vec{x}_3, \vec{x}_4\}$ represented by $\vec{v}_1 = (2.00, 1.75)$

    Cluster 2 : $\{\vec{x}_5, \vec{x}_6\}$ represented by $\vec{v}_2 = (2.00, 4.75)$

13.2.3 The algorithm

Notations

We assume that each data point is a n-dimensional vector:

$$\vec{x} = (x_1, x_2, \ldots, x_n).$$

The distance between two data points

$$\vec{x} = (x_1, x_2, \ldots, x_n)$$

and

$$\vec{y} = (y_1, y_2, \ldots, x_n)$$

is defined as

$$\|\vec{x} - \vec{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots (x_n - y_n)^2}.$$

Let $X = \{\vec{x}_1, \ldots, \vec{x}_N\}$ be the set of data points, $V = \{\vec{v}_1, \ldots, \vec{v}_k\}$ be the set of centres and c_i for $i = 1, \ldots, k$ be the number of data points in the i-th cluster

Basic idea

What the algorithm aims to achieve is to find a partition the set X into k mutually disjoint subsets $S = \{S_1, S_2, \ldots, S_k\}$ and a set of data points V which minimizes the following within-cluster sum of errors:

$$\sum_{i=1}^{k} \sum_{\vec{x} \in S_i} \|\vec{x} - \vec{v}_i\|^2$$

Algorithm

Step 1. Randomly select k cluster centers $\vec{v}_1, \ldots, \vec{v}_k$.

Step 2. Calculate the distance between each data point $\vec{x}_i$ and each cluster center $\vec{v}_j$.

Step 3. For each $j = 1, 2, \ldots, N$, assign the data point $\vec{x}_j$ to the cluster center $\vec{v}_i$ for which the distance $\|\vec{x}_j - \vec{v}_i\|$ is minimum. Let $\vec{x}_{i1}, \vec{x}_{i2}, \ldots, \vec{x}_{ic_i}$ be the data points assigned to $\vec{v}_i$.

Step 4. Recalculate the cluster centres using

$$\vec{v}_i = \frac{1}{c_i}(\vec{x}_{i1} + \cdots + \vec{x}_{ic_i}), \quad i = 1, 2, \ldots, k.$$

Step 5. Recalculate the distance between each data point and newly obtained cluster centers.

Step 6. If no data point was reassigned then stop. Otherwise repeat from Step 3.

Some methods for initialisation

The following are some of the methods for choosing the initial v_i's.

- Randomly take some k data points as the initial v_i's.
- Calculate the mean of all data and add small random vectors to the mean to get the k initial v_i's.
- Calculate the principal component, divide its range into k equal intervals, partition the data into k groups, and then take the means of these groups as the initial centres.

13.2.4 Disadvantages

Even though the k-means algorithm is fast, robust and easy to understand, there are several disadvantages to the algorithm.

- The learning algorithm requires apriori specification of the number of cluster centers.
- The final cluster centres depend on the initial v_i's.
- With different representation of data we get different results (data represented in form of cartesian co-ordinates and polar co-ordinates will give different results).
- Euclidean distance measures can unequally weight underlying factors.
- The learning algorithm provides the local optima of the squared error function.
- Randomly choosing of the initial cluster centres may not lead to a fruitful result.
- The algorithm cannot be applied to categorical data.

13.2.5 Application: Image segmentation and compression

Image segmentation

The goal of segmentation is to partition an image into regions each of which has a reasonably homogeneous visual appearance or which corresponds to objects or parts of objects. Each pixel in an image is a point in a 3-dimensional space comprising the intensities of the red, blue, and green channels. A segmentation algorithm simply treats each pixel in the image as a separate data point. For any value of k, each pixel is replaced by the pixel vector with the (R, G, B) intensity triplet given by the centre μ_k to which that pixel has been assigned. For a given value of k, the algorithm is representing the image using a palette of only k colours. It should be emphasized that this use of k-means is a very crude approach to image segmentation. The image segmentation problem is in general extremely difficult.

Data compression

We can also the clustering algorithm to perform data compression. There are two types of data compression: *lossless data compression*, in which the goal is to be able to reconstruct the original data exactly from the compressed representation, and *lossy data compression*, in which we accept some errors in the reconstruction in return for higher levels of compression than can be achieved in the lossless case.

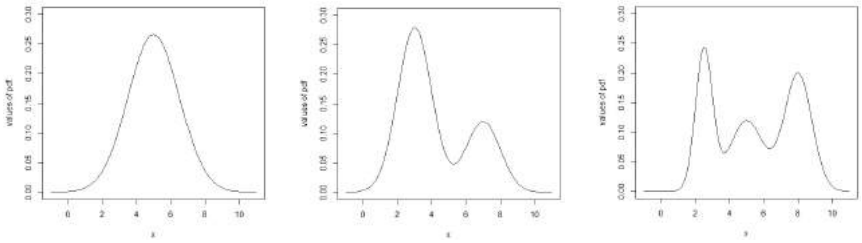
We can apply the k-means algorithm to the problem of lossy data compression as follows. For each of the N data points, we store only the identity of the cluster to which it is assigned. We also store the values of the k cluster centres μ_k, which requires much less data, provided we choose k much smaller than N. Each data point is then approximated by its nearest centre μ_k. New data points can similarly be compressed by first finding the nearest μ_k and then storing the label k instead of the original data vector. This framework is often called *vector quantization*, and the vectors Îijμ_k are called *code-book vectors*.

13.3 Multi-modal distributions

13.3.1 Definitions

1. In statistics, a *unimodal distribution* is a continuous probability distribution with only one mode (or "peak").

   A random variable having the normal distribution is a unimodal distribution. Similarly, the t-distribution and the chi-squared distribution are also unimodal distributions.



Unimodal Bimodal Multimodal

Figure 13.5: Probability distributions

2. A *bimodal distribution* is a continuous probability distribution with two different modes. The modes appear as distinct peaks in the graph of the probability density function.

3. A *multimodal distribution* is a continuous probability distribution with two or more modes.

13.4 Mixture of normal distributions

13.4.1 Bimodal mixture

Consider the following functions which are probability density functions of normally distributed random variables.

$$f_1(x) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} \tag{13.1}$$

$$f_2(x) = \frac{1}{\sigma_2\sqrt{2\pi}} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}} \tag{13.2}$$

Now consider the following function:

$$f(x) = \pi_1 f_1(x) + \pi_2 f_2(x) \tag{13.3}$$

where π_1 and π_2 are some constants satisfying the relation

$$\pi_1 + \pi_2 = 1. \tag{13.4}$$

It can be shown that the function given in Eq.(13.3) together with Eq.(13.4) defines a probability density function. It can also be shown that the graph of this function has two peaks. Hence this function defines a bimodal distribution. This distribution is called a mixture of the normal distributions defined by Eqs.(13.1) and (13.2). We may mix more than two normal distributions.

13.4.2 Definition

Consider the following k probability density functions:

$$f_i(x) = \frac{1}{\sigma_i\sqrt{2\pi}} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}, \quad i = 1, 2, \ldots, k. \tag{13.5}$$

Let $\pi_1, \pi_2, \ldots, \pi_k$ be constants such that

$$\pi_i \geq 0, \quad i = 1, 2, \ldots, k \tag{13.6}$$

$$\pi_1 + \pi_2 + \cdots + \pi_k = 1. \tag{13.7}$$

Then the random variable X whose probability density function is

$$f(x) = f_1(x) + f_2(x) + \cdots + f_k(x), \tag{13.8}$$

is said to be a *mixture of the k normal distributions* having the probability density functions defined in Eq.(13.5).

A natural example

As a natural example for such mixtures of normal populations, we consider the probability distribution of heights of people in a region. This is a mixture of two normal distributions: the distribution of heights of males and the distribution of heights of females. Given only the height data and not the gender assignments for each data point, the distribution of all heights would follow the weighted sum of two normal distributions.

13.4.3 Example for mixture of two normal distributions

Data and histogram

Consider the 100 observations of some attribute X given in Table 13.2.

```
 [1] 5.39 1.30 2.95 2.16 2.37 2.33 4.76 2.99 1.71 2.41
[11] 2.71 2.79 0.54 1.37 5.16 1.22 1.58 4.34 3.83 3.44
[21] 3.68 5.03 0.92 2.57 1.97 2.17 5.02 2.73 1.63 3.09
[31] 4.05 3.76 3.13 6.50 5.10 3.62 3.14 2.36 2.73 4.08
[41] 3.28 2.28 1.52 3.86 2.10 0.86 2.94 2.18 3.39 2.55
[51] 3.23 3.30 2.16 3.86 1.92 2.55 4.33 0.86 2.68 2.24
[61] 2.82 3.63 2.84 3.82 2.49 3.25 2.39 3.18 6.35 4.16
[71] 6.68 5.26 8.00 6.27 7.98 6.50 6.56 8.50 7.48 6.42
[81] 5.99 7.44 6.96 7.10 8.48 6.99 7.29 6.87 6.71 7.99
[91] 8.19 8.28 6.98 7.43 8.33 5.65 8.96 7.36 5.24 7.30
```

Table 13.2: A set of 100 observations of a numeric attribute X

To make some sense of this set of observations, let us construct the frequency table for the data as in Table 13.3.

Range	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7 -8	8-9	9-10
Frequency	4	9	26	18	6	9	12	9	7	0
Relative frequency	0.04	0.09	0.26	0.18	0.06	0.09	0.12	0.09	0.07	0.00

Table 13.3: Frequency table of data in Table 13.2

Figure 13.6 shows the histogram of the relative frequencies. Notice that the histogram has two "peaks", one near $x = 2.5$ and one near $x = 6.5$. So, the graph of the probability density function of the attribute X must have two peaks. Recall that the graph of the probability density function of a random variable having the normal distribution has only one peak.

Probability distribution

The data in Table 13.2 was generated using the R programming language. It is a true "mixture" of the values two normally distributed random variables. 70% of the observations are random values of a normally distributed random variable with $\mu_1 = 3$ and $\sigma_1 = 1.20$ and 30% of the observations are values of a normally distributed random variable with $\mu_2 = 7$ and $\sigma_2 = 0.87$. The weight for the first normal distribution is $\pi_1 = 70\% = 0.7$ and that for the second distribution is $\pi_2 = 30\% = 0.3$. The probability density function for the mixed distribution is

$$f(x) = 0.7 \times \frac{1}{1.20\sqrt{2\pi}} e^{-(x-3)^2/(2\times 1.20^2)} + 0.3 \times \frac{1}{0.87\sqrt{2\pi}} e^{-(x-7)^2/(2\times 0.87^2)}. \tag{13.9}$$

Figure 13.6 also shows the curve defined by Eq.(13.9) superimposed on the histogram of the relative frequency distribution.

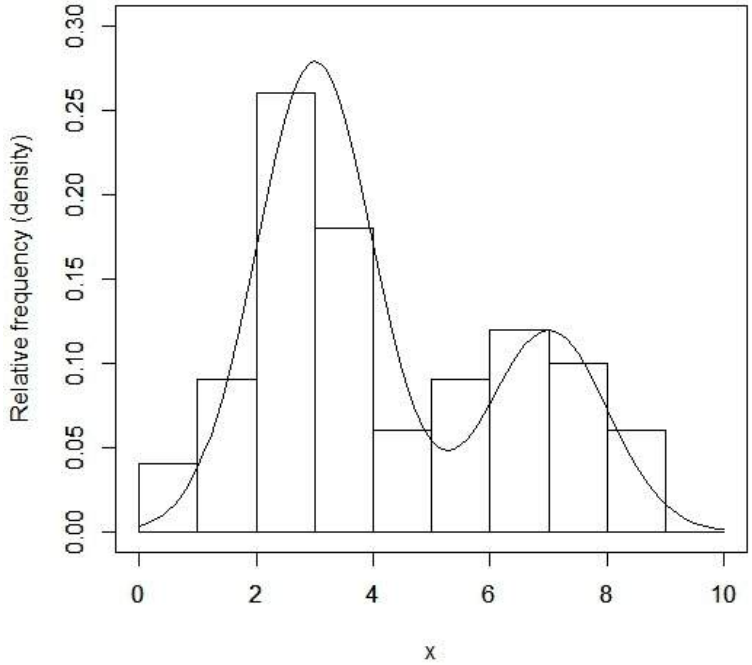


Figure 13.6: Graph of pdf defined by Eq.(13.9) superimposed on the histogram of the data in Table 13.3

13.5 Mixtures in terms of latent variables

Consider the mixture of k normal distributions defined by Eqs.(13.5) – (13.8).

Let us define a k-dimensional random variable

$$\vec{Z} = (z_1, z_2, \ldots, z_k)$$

where each z_1 is either 0 or 1 and a 1 appears only at one place; that is,

$$z_i \in \{0, 1\} \text{ and } z_1 + z_2 + \cdots + z_k = 0.$$

We also assume that

$$P(z_k = 1) = \pi_k.$$

The probability function of $\vec{Z}$ can be written in the form

$$P(\vec{Z}) = \pi_1^{z_1} \pi_2^{z_2} \ldots \pi_k^{z_k}.$$

Now, suppose we have a set of observations $\{x_1, x_2, \ldots, x_N\}$. Suppose that, in some way, we can associate a value of the random variable $\vec{Z}$, say $\vec{Z}_i$, with each value x_i and think of the given set of observations as a set of ordered pairs

$$\{(x_1, \vec{Z}_1), (x_2, \vec{Z}_2), \ldots, (x_N, \vec{Z}_N)\}.$$

Here, only the x_i-s are known; the $\vec{Z}_i$-s are unknown. Let us further assume that the conditional probability distribution $p(x|\vec{Z})$ be given by

$$p(x|\vec{Z}) = [f_1(x)]^{z_1} \times \cdots \times [f_k(x)]^{z_k}.$$

Then the marginal distribution of x is given by

$$\begin{aligned} p(x) &= \sum_{\vec{Z}} p(\vec{Z}) P(x|\vec{Z}) \\ &= \pi_1 f_1(x) + \cdots + \pi_k f_k(x). \end{aligned} \tag{13.10}$$

The right hand side of Eq.(13.10) is the probability density function of a mixture of k normal distributions with weights $\pi_1, \ldots, \pi_k$.

Thus, a mixture of normal distributions is the marginal distribution of a bivariate distribution $(x, \vec{Z})$ where $\vec{Z}$ is an unobserved or latent variable.

13.6 Expectation-maximisation algorithm

The maximum likelihood estimation method (MLE) is a method for estimating the parameters of a statistical model, given observations (see Section 6.5 for details). The method attempts to find the parameter values that maximize the likelihood function, or equivalently the log-likelihood function, given the observations.

The *expectation-maximisation algorithm* (sometimes abbreviated as the *EM algorithm*) is used to find maximum likelihood estimates of the parameters of a statistical model in cases where the equations cannot be solved directly. These models generally involve latent or unobserved variables in addition to unknown parameters and known data observations. For example, a Gaussian mixture model can be described by assuming that each observed data point has a corresponding unobserved data point, or latent variable, specifying the mixture component to which each data point belongs.

The EM Algorithm is not really an algorithm. Rather it is a general procedure to create algorithms for specific MLE problems. The complete details of this general procedure are beyond the scope of this book. However, we present below a minimal outline of the algorithm

Outline of EM algorithm

Step 1. Initialise the parameters θ to be estimated.

Step 2. **Expectation step (E-step)**

Take the expected value of the complete data given the observation and the current parameter estimate, say, $\hat{\theta}_j$. This is a function of θ and $\hat{\theta}_j$, say, $Q(\theta, \hat{\theta}_j)$.

Step 3. **Maximization step (M-step)**

Find the values θ that maximizes the function $Q(\theta, \hat{\theta}_j)$.

Step 4. Repeat Steps 1 and 2 until the parameter values or the likelihood function converge.

13.7 The EM algorithm for Gaussian mixtures

In the case of Gaussian mixture problems, because of the nature of the function, finding a maximum likelihood estimate by taking the derivatives of the log-likelihood function with respect to all the parameters and simultaneously solving the resulting equations is nearly impossible. So we apply the EM algorithm to solve the problem.

As already indicated, the EM algorithm is a general procedure for estimating the parameters in a statistical model. This algorithm can be adapted to develop an algorithm for estimating the parameters in a Gaussian mixture model. The adapted EM algorithm has been explained below. (The details of how the EM algorithm can be adapted to estimate the parameters in a Gaussian mixture model are also beyond the scope of this book. For details on these matters, one may refer to [1]).

Problem

Suppose we are given a set of N observations

$$\{x_1, x_2, \ldots, x_N\}$$

of a numeric variable X. Let X be a mix of k normal distributions and let the probability density function of X be

$$f(x) = \pi_1 f_1(x) + \cdots + \pi_k f_k(x)$$

where

$$\pi_i \geq 0, \quad i = 1, 2, \ldots, k$$
$$\pi_i + \cdots + \pi_k = 1$$
$$f_i(x) = \frac{1}{\sigma_i \sqrt{2\pi}} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}, \quad i = 1, 2, \ldots, k.$$

Estimate the parameters $\mu_1, \ldots, \mu_k$, $\sigma_1, \ldots, \sigma_k$ and $\pi_1 \ldots, \pi_k$.

Log-likelihood function

Let θ denote the set of parameters $\mu_i, \sigma_i, \pi_i (i = 1, \ldots, k)$. The log-likelihood function for the above problem is given below:

$$\begin{aligned} \log L(\theta) &= \log f(x_1) + \cdots + \log f(x_N) \\ &= \sum_{i=1}^{N} \log \left(\frac{\pi_1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(x_i-\mu_1)^2}{2\sigma_1^2}} + \cdots + \frac{\pi_k}{\sigma_k \sqrt{2\pi}} e^{-\frac{(x_i-\mu_k)^2}{2\sigma_k^2}} \right) \end{aligned} \tag{13.11}$$

The algorithm

Step 1. Initialise the means μ_i's, the variances σ_i^2's and the mixing coefficients π_i's.

Step 2. Calculate the following for $n = 1, \ldots, N$ and $i = 1, \ldots, k$:

$$\gamma_{in} = \frac{\pi_i f_i(x_n)}{\pi_1 f_1(x_n) + \cdots + \pi_k f_k(x_n)}$$
$$N_i = \gamma_{i1} + \cdots + \gamma_{iN}$$

Step 3. Recalculate the parameters using the following:

$$\mu_i^{(\text{new})} = \frac{1}{N_i} \left(\gamma_{i1} x_1 + \cdots \gamma_{iN} x_N \right)$$

$$\sigma_i^{2(\text{new})} = \frac{1}{N_i}\left(\gamma_{i1}(x_1 - \mu_i^{(\text{new})})^2 + \cdots + \gamma_{iN}(x_1 - \mu_i^{(\text{new})})^2\right)$$
$$\pi_i^{(\text{new})} = \frac{N_i}{N}$$

Step 4. Evaluate the log-likelihood function given in Eq.(13.11) and check for convergence of either the parameters or the log-likelihood function. If the convergence criterion is not satisfied, return to Step 2.

13.8 Hierarchical clustering

Hierarchical clustering (also called hierarchical cluster analysis or HCA) is a method of cluster analysis which seeks to build a hierarchy of clusters (or groups) in a given dataset. The hierarchical clustering produces clusters in which the clusters at each level of the hierarchy are created by merging clusters at the next lower level. At the lowest level, each cluster contains a single observation. At the highest level there is only one cluster containing all of the data.

The decision regarding whether two clusters are to be merged or not is taken based on the *measure of dissimilarity* between the clusters. The distance between two clusters is usually taken as the measure of dissimilarity between the clusters.

In Section **??**, we shall see various methods for measuring the distance between two clusters.

13.8.1 Dendrograms

Hierarchical clustering can be represented by a rooted binary tree. The nodes of the trees represent groups or clusters. The root node represents the entire data set. The terminal nodes each represent one of the individual observations (singleton clusters). Each nonterminal node has two daughter nodes.

The distance between merged clusters is monotone increasing with the level of the merger. The height of each node above the level of the terminal nodes in the tree is proportional to the value of the distance between its two daughters (see Figure 13.9).

A *dendrogram* is a tree diagram used to illustrate the arrangement of the clusters produced by hierarchical clustering.

The dendrogram may be drawn with the root node at the top and the branches growing vertically downwards (see Figure 13.8(a)). It may also be drawn with the root node at the left and the branches growing horizontally rightwards (see Figure 13.8(b)). In some contexts, the opposite directions may also be more appropriate.

Dendrograms are commonly used in computational biology to illustrate the clustering of genes or samples.

Example

Figure 13.7 is a dendrogram of the dataset $\{a, b, c, d, e\}$. Note that the root node represents the entire dataset and the terminal nodes represent the individual observations. However, the dendrograms are presented in a simplified format in which only the terminal nodes (that is, the nodes representing the singleton clusters) are explicitly displayed. Figure 13.8 shows the simplified format of the dendrogram in Figure 13.7.

Figure 13.9 shows the distances of the clusters at the various levels. Note that the clusters are at 4 levels. The distance between the clusters $\{a\}$ and $\{b\}$ is 15, between $\{c\}$ and $\{d\}$ is 7.5, between $\{c, d\}$ and $\{e\}$ is 15 and between $\{a, b\}$ and $\{c, d, e\}$ is 25.

13.8.2 Methods for hierarchical clustering

There are two methods for the hierarchical clustering of a dataset. These are known as the *agglomerative method* (or the bottom-up method) and the *divisive method* (or, the top-down method).

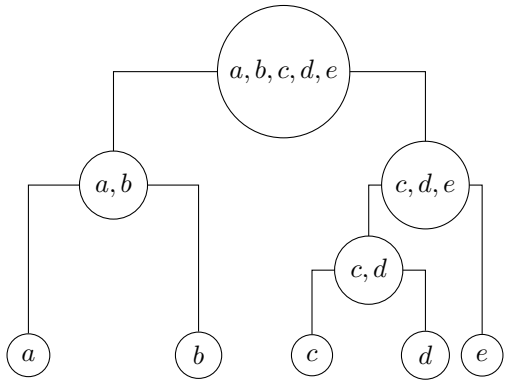


Figure 13.7: A dendrogram of the dataset $\{a, b, c, d, e\}$

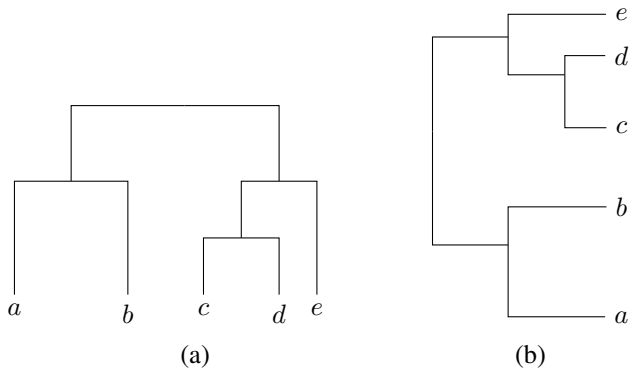


Figure 13.8: Different ways of drawing dendrogram

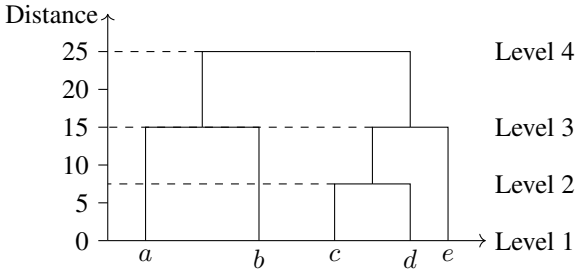


Figure 13.9: A dendrogram of the dataset $\{a, b, c, d, e\}$ showing the distances (heights) of the clusters at different levels

Agglomerative method

In the agglomerative we start at the bottom and at each level recursively merge a selected pair of clusters into a single cluster. This produces a grouping at the next higher level with one less cluster. If there are N observations in the dataset, there will be $N-1$ levels in the hierarchy. The pair chosen for merging consist of the two groups with the smallest "intergroup dissimilarity".

For example, the hierarchical clustering shown in Figure 13.7 can be constructed by the agglomerative method as shown in Figure 13.10. Each nonterminal node has two daughter nodes. The daughters represent the two groups that were merged to form the parent.

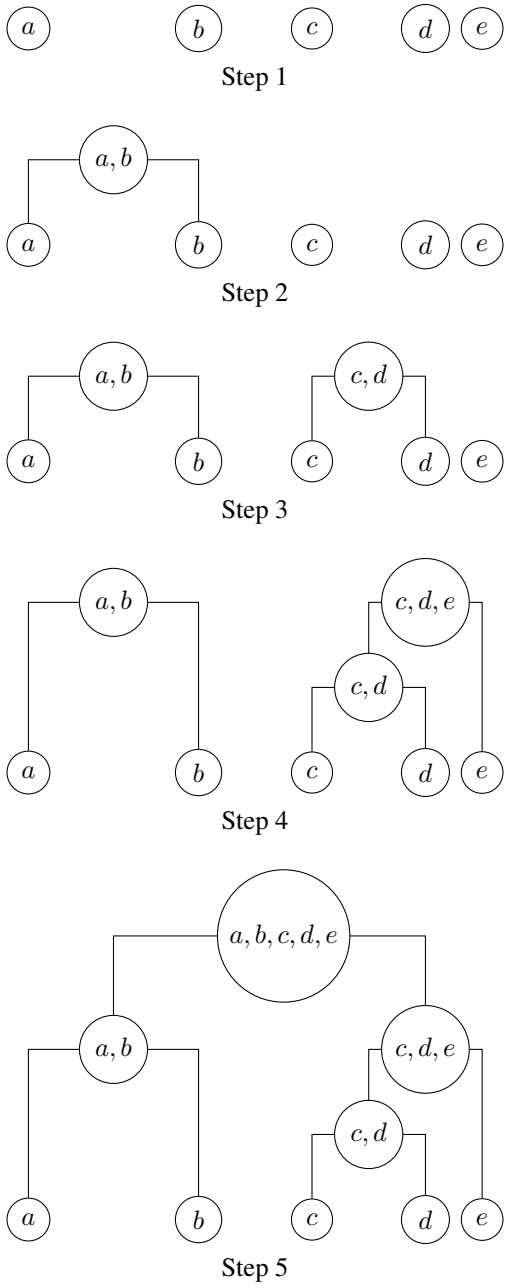


Figure 13.10: Hierarchical clustering using agglomerative method

Divisive method

The divisive method starts at the top and at each level recursively split one of the existing clusters at that level into two new clusters. If there are N observations in the dataset, there the divisive method also will produce $N - 1$ levels in the hierarchy. The split is chosen to produce two new groups with the largest "between-group dissimilarity".

For example, the hierarchical clustering shown in Figure 13.7 can be constructed by the divisive method as shown in Figure 13.11. Each nonterminal node has two daughter nodes. The two daughters represent the two groups resulting from the split of the parent.

13.9 Measures of dissimilarity

In order to decide which clusters should be combined (for agglomerative), or where a cluster should be split (for divisive), a measure of dissimilarity between sets of observations is required. In most methods of hierarchical clustering, the dissimilarity between two groups of observations is measured by using an appropriate measure of distance between the groups of observations. The distance between two groups of observations is defined in terms of the distance between two observations. There are several ways in which the distance between two observations can be defined and also there are also several ways in which the distance between two groups of observations can be defined.

13.9.1 Measures of distance between data points

Numeric data

We assume that each observation or data point is a n-dimensional vector. Let $\vec{x} = (x_1, \ldots, x_n)$ and $\vec{y} = (y_1, \ldots, y_n)$ be two observations. Then the following are the commonly used measures of distances in the hierarchical clustering of numeric data.

Name	Formula
Euclidean distance	$\|\vec{x} - \vec{y}\|_2 = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$
Squared Euclidean distance	$\|\vec{x} - \vec{y}\|_2^2 = (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2$
Manhattan distance	$\|\vec{x} - \vec{y}\|_1 = \|x_1 - y_1\| + \cdots + \|x_n - y_n\|$
Maximum distance	$\|\vec{x} - \vec{y}\|_\infty = \max\{\|x_1 - y_1\|, \ldots, \|x_n - y_n\|\}$

Non-numeric data

For text or other non-numeric data, metrics such as the Levenshtein distance are often used.

The *Levenshtein distance* is a measure of the "distance" between two words. The Levenshtein distance between two words is the minimum number of single-character edits (insertions, deletions or substitutions) required to change one word into the other.

For example, the Levenshtein distance between "kitten" and "sitting" is 3, since the following three edits change one into the other, and there is no way to do it with fewer than three edits:

kitten → sitten (substitution of "s" for "k")
sitten → sittin (substitution of "i" for "e")
sittin → sitting (insertion of"g" at the end)

13.9.2 Measures of distance between groups of data points

Let A and B be two groups of observations and let x and y be arbitrary data points in A and B respectively. Suppose we have chosen some formula, say Euclidean distance formula, to measure the distance between data points. Let $d(x, y)$ denote the distance between x and y. We denote by

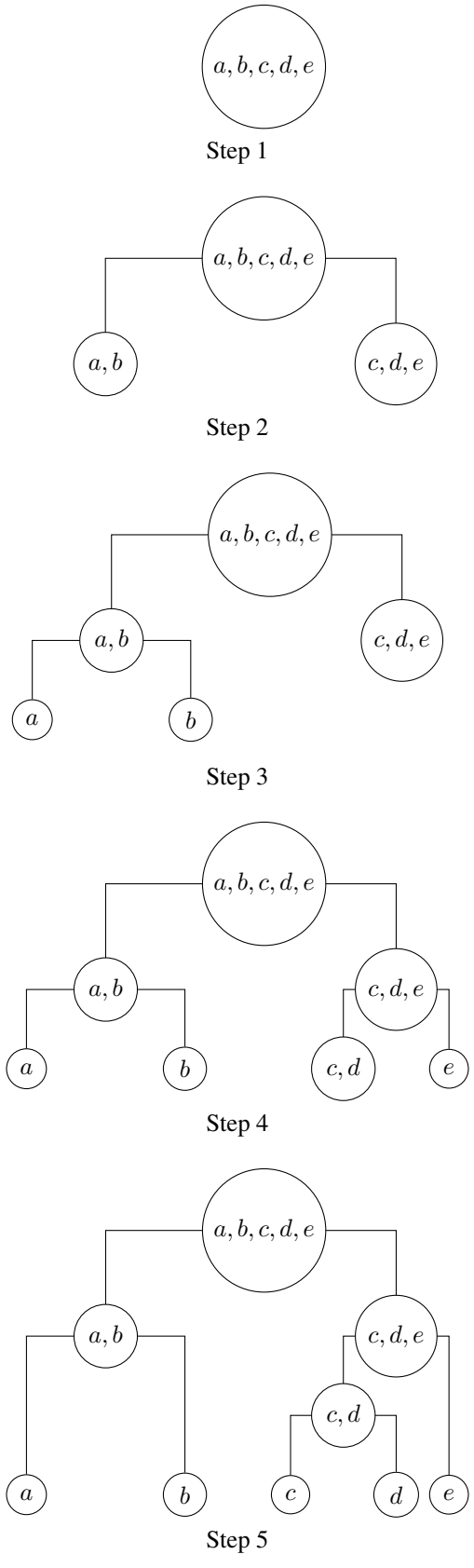


Figure 13.11: Hierarchical clustering using divisive method

$d(A, B)$ the distance between the groups A and B. The following are some of the different methods in which $d(A, B)$ is defined.

1. $d(A, B) = \max\{d(x, y) : x \in A, y \in B\}$.

   Agglomerative hierarchical clustering using this measure of dissimilarity is known as *complete-linkage clustering*. The method is also known as *farthest neighbour clustering*.

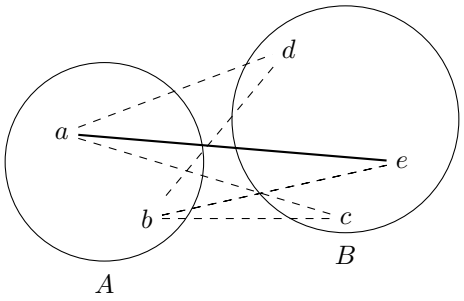


Figure 13.12: Length of the solid line "ae" is $\max\{d(x, y) : x \in A, y \in B\}$

2. $d(A, B) = \min\{d(x, y) : x \in A, y \in B\}$.

   Agglomerative hierarchical clustering using this measure of dissimilarity is known as *single-linkage clustering*. The method is also known as *nearest neighbour clustering*.

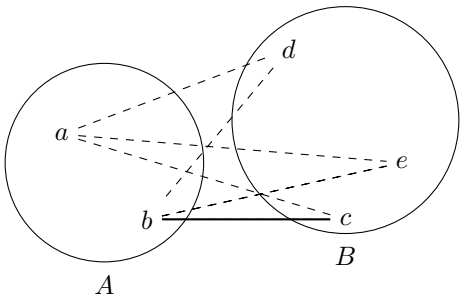


Figure 13.13: Length of the solid line "bc" is $\min\{d(x, y) : x \in A, y \in B\}$

3. $d(A, B) = \dfrac{1}{|A||B|} \displaystyle\sum_{x \in A, y \in B} d(x, y)$ where $|A|$, $|B|$ are respectively the number of elements in A and B.

   Agglomerative hierarchical clustering using this measure of dissimilarity is known as *mean or average linkage clustering*. It is also known as UPGMA (Unweighted Pair Group Method with Arithmetic Mean).

13.10 Algorithm for agglomerative hierarchical clustering

Given a set of N items to be clustered and an $N \times N$ distance matrix, required to construct a hierarchical clustering of the data using the agglomerative method.

Step 1. Start by assigning each item to its own cluster, so that we have N clusters, each containing just one item. Let the distances between the clusters equal the distances between the items they contain.

Step 2. Find the closest pair of clusters and merge them into a single cluster, so that now we have one less cluster.

Step 3. Compute distances between the new cluster and each of the old clusters.

Step 4. Repeat Steps 2 and 3 until all items are clustered into a single cluster of size N.

13.10.1 Example

Problem 1

Given the dataset $\{a, b, c, d, e\}$ and the following distance matrix, construct a dendrogram by complete-linkage hierarchical clustering using the agglomerative method.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Table 13.4: Example for distance matrix

Solution

The complete-linkage clustering uses the "maximum formula", that is, the following formula to compute the distance between two clusters A and B:

$$d(A, B) = \max\{d(x, y) : x \in A, y \in B\}$$

1. Dataset : $\{a, b, c, d, e\}$.

   Initial clustering (singleton sets) C_1: $\{a\}$, $\{b\}$, $\{c\}$, $\{d\}$, $\{e\}$.

2. The following table gives the distances between the various clusters in C_1:

	$\{a\}$	$\{b\}$	$\{c\}$	$\{d\}$	$\{e\}$
$\{a\}$	0	9	3	6	11
$\{b\}$	9	0	7	5	10
$\{c\}$	3	7	0	9	**2**
$\{d\}$	6	5	9	0	8
$\{e\}$	11	10	**2**	8	0

   In the above table, the minimum distance is the distance between the clusters $\{c\}$ and $\{e\}$. Also

   $$d(\{c\}, \{e\}) = 2.$$

   We merge $\{c\}$ and $\{e\}$ to form the cluster $\{c, e\}$.

   The new set of clusters C_2: $\{a\}$, $\{b\}$, $\{d\}$, $\{c, e\}$.

3. Let us compute the distance of $\{c, e\}$ from other clusters.

   $d(\{c, e\}, \{a\}) = \max\{d(c, a), d(e, a)\} = \max\{3, 11\} = 11.$

   $d(\{c, e\}, \{b\}) = \max\{d(c, b), d(e, b)\} = \max\{7, 10\} = 10.$

   $d(\{c, e\}, \{d\}) = \max\{d(c, d), d(e, d)\} = \max\{9, 8\} = 9.$

   The following table gives the distances between the various clusters in C_2.

	$\{a\}$	$\{b\}$	$\{d\}$	$\{c,e\}$
$\{a\}$	0	9	6	11
$\{b\}$	9	0	**5**	10
$\{d\}$	6	**5**	0	9
$\{c,e\}$	11	10	9	0

In the above table, the minimum distance is the distance between the clusters $\{b\}$ and $\{d\}$. Also

$$d(\{b\},\{d\}) = 5.$$

We merge $\{b\}$ and $\{d\}$ to form the cluster $\{b,d\}$.

The new set of clusters C_3: $\{a\}$, $\{b,d\}$, $\{c,e\}$.

4. Let us compute the distance of $\{b,d\}$ from other clusters.

   $d(\{b,d\},\{a\}) = \max\{d(b,a),d(d,a)\} = \max\{9,6\} = 9$.

   $d(\{b,d\},\{c,e\}) = \max\{d(b,c),d(b,e),d(d,c),d(d,e)\} = \max\{7,10,9,8\} = 10$.

   The following table gives the distances between the various clusters in C_3.

	$\{a\}$	$\{b,d\}$	$\{c,e\}$
$\{a\}$	0	**9**	11
$\{b,d\}$	**9**	0	10
$\{c,e\}$	11	10	0

   In the above table, the minimum distance is the distance between the clusters $\{a\}$ and $\{b,d\}$. Also

$$d(\{a\},\{b,d\}) = 9.$$

   We merge $\{a\}$ and $\{b,d\}$ to form the cluster $\{a,b,d\}$.

   The new set of clusters C_4: $\{a,b,d\}$, $\{c,e\}$

5. Only two clusters are left. We merge them form a single cluster containing all data points. We have

$$\begin{aligned} d(\{a,b,d\},\{c,e\}) &= \max\{d(a,c),d(a,e),d(b,c),d(b,e),d(d,c),d(d,e)\} \\ &= \max\{3,11,7,10,9,8\} \\ &= 11 \end{aligned}$$

6. Figure 13.14 shows the dendrogram of the hierarchical clustering.

Problem 2

Given the dataset $\{a,b,c,d,e\}$ and the distance matrix given in Table 13.4, construct a dendrogram by single-linkage hierarchical clustering using the agglomerative method.

Solution

The complete-linkage clustering uses the "maximum formula", that is, the following formula to compute the distance between two clusters A and B:

$$d(A,B) = \min\{d(x,y) : x \in A, y \in B\}$$

1. Dataset : $\{a,b,c,d,e\}$.

   Initial clustering (singleton sets) C_1: $\{a\}$, $\{b\}$, $\{c\}$, $\{d\}$, $\{e\}$.

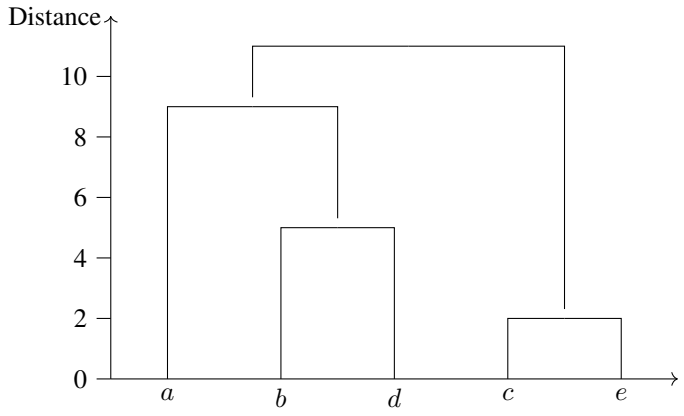


Figure 13.14: Dendrogram for the data given in Table 13.4 (complete linkage clustering)

2. The following table gives the distances between the various clusters in C_1:

	$\{a\}$	$\{b\}$	$\{c\}$	$\{d\}$	$\{e\}$
$\{a\}$	0	9	3	6	11
$\{b\}$	9	0	7	5	10
$\{c\}$	3	7	0	9	**2**
$\{d\}$	6	5	9	0	8
$\{e\}$	11	10	**2**	8	0

In the above table, the minimum distance is the distance between the clusters $\{c\}$ and $\{e\}$. Also

$$d(\{c\},\{e\}) = 2.$$

We merge $\{c\}$ and $\{e\}$ to form the cluster $\{c,e\}$.

The new set of clusters C_2: $\{a\}$, $\{b\}$, $\{d\}$, $\{c,e\}$.

3. Let us compute the distance of $\{c,e\}$ from other clusters.

$d(\{c,e\},\{a\}) = \min\{d(c,a),d(e,a)\} = \max\{3,11\} = 3.$

$d(\{c,e\},\{b\}) = \min\{d(c,b),d(e,b)\} = \max\{7,10\} = 7.$

$d(\{c,e\},\{d\}) = \min\{d(c,d),d(e,d)\} = \max\{9,8\} = 8.$

The following table gives the distances between the various clusters in C_2.

	$\{a\}$	$\{b\}$	$\{d\}$	$\{c,e\}$
$\{a\}$	0	9	6	**3**
$\{b\}$	9	0	5	7
$\{d\}$	6	5	0	8
$\{c,e\}$	**3**	7	8	0

In the above table, the minimum distance is the distance between the clusters $\{a\}$ and $\{c,e\}$. Also

$$d(\{a\},\{c,e\}) = 3.$$

We merge $\{a\}$ and $\{c,e\}$ to form the cluster $\{a,c,e\}$.

The new set of clusters C_3: $\{a,c,e\}$, $\{b\}$, $\{d\}$.

4. Let us compute the distance of $\{a, c, e\}$ from other clusters.
   $d(\{a, c, e\}, \{b\})$ = $\min\{d(a, b), d(c, b), d(e, b)\}$ = $\{9, 7, 10\}$ = 7
   $d(\{a, c, e\}, \{d\})$ = $\min\{d(a, d), d(c, d), d(e, d)\}$ = $\{6, 9, 8\}$ = 6
   The following table gives the distances between the various clusters in C_3.

	$\{a, c, e\}$	$\{b\}$	$\{d\}$
$\{a, c, e\}$	0	7	6
$\{b\}$	7	0	**5**
$\{d\}$	6	**5**	0

   In the above table, the minimum distance is between $\{b\}$ and $\{d\}$. Also

$$d(\{b\}, \{d\}) = 5.$$

   We merge $\{b\}$ and $\{d\}$ to form the cluster $\{b, d\}$.
   The new set of clusters C_4: $\{a, c, e\}$, $\{b, d\}$

5. Only two clusters are left. We merge them form a single cluster containing all data points. We have

$$\begin{aligned} d(\{a, c, e\}, \{b, d\}) &= \min\{d(a, b), d(a, d), d(c, b), d(c, d), d(e, b), d(e, d)\} \\ &= \min\{9, 6, 7, 9, 10, 8\} \\ &= 6 \end{aligned}$$

6. Figure 13.15 shows the dendrogram of the hierarchical clustering.

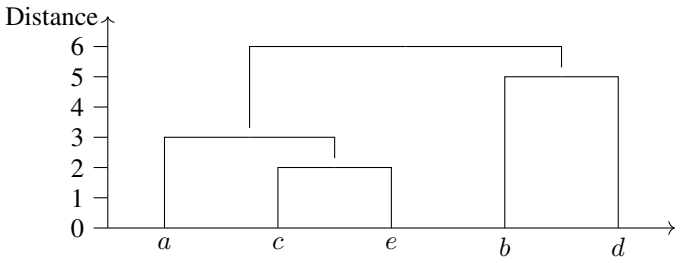


Figure 13.15: Dendrogram for the data given in Table 13.4 (single linkage clustering)

13.11 Algorithm for divisive hierarchical clustering

Divisive clustering algorithms begin with the entire data set as a single cluster, and recursively divide one of the existing clusters into two daughter clusters at each iteration in a top-down fashion. To apply this procedure, we need a separate algorithm to divide a given dataset into two clusters.

- The divisive algorithm may be implemented by using the k-means algorithm with k = 2 to perform the splits at each iteration. However, it would not necessarily produce a splitting sequence that possesses the monotonicity property required for dendrogram representation.

13.11.1 DIANA (DIvisive ANAlysis)

DIANA is a divisive hierarchical clustering technique. Here is an outline of the algorithm.

Step 1. Suppose that cluster C_l is going to be split into clusters C_i and C_j.

Step 2. Let $C_i = C_l$ and $C_j = \varnothing$.

Step 3. For each object $x \in C_i$:

(a) For the first iteration, compute the average distance of x to all other objects.

(b) For the remaining iterations, compute

$$D_x = \text{average}\,\{d(x,y) : y \in C_i\} - \text{average}\{d(x,y) : y \in C_j\}.$$

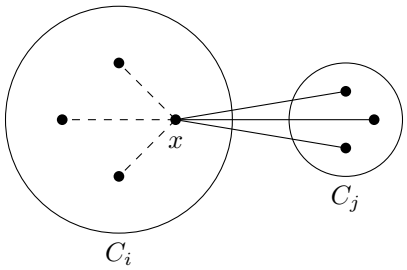


Figure 13.16: D_x= (average of dashed lines) − (average of solid lines)

Step 4. (a) For the first iteration, move the object with the maximum average distance to C_j.

(b) For the remaining iterations, find an object x in C_i for which D_x is the largest. If $D_x > 0$ then move x to C_j.

Step 5. Repeat Steps 3(b) and 4(b) until all differences D_x are negative. Then C_l is split into C_i and C_j.

Step 6. Select the smaller cluster with the largest diameter. (The diameter of a cluster is the largest dissimilarity between any two of its objects.) Then divide this cluster, following Steps 1-5.

Step 7. Repeat Step 6 until all clusters contain only a single object.

13.11.2 Example

Problem

Given the dataset $\{a, b, c, d, e\}$ and the distance matrix in Table 13.4, construct a dendrogram by the divisive analysis algorithm.

Solution

1. We have, initially
$$C_l = \{a, b, c, d, e\}$$
2. We write
$$C_i = C_l, \quad C_j = \varnothing.$$
3. Division into clusters

(a) Initial iteration

Let us calculate the average dissimilarities of the objects in C_i with the other objects in C_i.

Average dissimilarity of a

$$= \frac{1}{4}(d(a,b) + d(a,c) + d(a,e)) = \frac{1}{4}(9 + 3 + 6 + 11) = 7.25$$

Similarly we have :

Average dissimilarity of b = 7.75

Average dissimilarity of c = 5.25

Average dissimilarity of d = 7.00

Average dissimilarity of e = 7.75

The highest average distance is 7.75 and there are two corresponding objects. We choose one of them, b, arbitrarily. We move b to C_j.

We now have

$$C_i = \{a, c, d, e\}, \quad C_j = \varnothing \cup \{b\} = \{b\}.$$

(b) Remaining iterations

(i) 2-nd iteration.

$$D_a = \frac{1}{3}(d(a,c) + d(a,d) + d(a,e)) - \frac{1}{1}(d(a,b)) = \frac{20}{3} - 9 = -2.33$$
$$D_c = \frac{1}{3}(d(c,a) + d(c,d) + d(c,e)) - \frac{1}{1}(d(c,b)) = \frac{14}{3} - 7 = -2.33$$
$$D_d = \frac{1}{3}(d(d,a) + d(d,c) + d(d,e)) - \frac{1}{1}(d(c,b)) = \frac{23}{3} - 7 = 0.67$$
$$D_e = \frac{1}{3}(d(e,a) + d(e,c) + d(e,d)) - \frac{1}{1}(d(e,b)) = \frac{21}{3} - 7 = 0$$

D_d is the largest and $D_d > 0$. So we move, d to C_j.

We now have

$$C_i = \{a, c, e\}, \quad C_j = \{b\} \cup \{d\} = \{b, d\}.$$

(ii) 3-rd iteration

$$D_a = \frac{1}{2}(d(a,c) + d(a,e)) - \frac{1}{2}(d(a,b) + d(a,d)) = \frac{14}{2} - \frac{15}{2} = -0.5$$
$$D_c = \frac{1}{2}(d(c,a) + d(c,e)) - \frac{1}{2}(d(c,b) + d(c,d)) = \frac{5}{2} - \frac{16}{2} = -13.5$$
$$D_e = \frac{1}{2}(d(e,a) + d(e,c)) - \frac{1}{2}(d(e,b) + d(e,d)) = \frac{13}{2} - \frac{18}{2} = -2.5$$

All are negative. So we stop and form the clusters C_i and C_j.

4. To divide, C_i and C_j, we compute their diameters.

$$\begin{aligned} \text{diameter}(C_i) &= \max\{d(a,c), d(a,e), d(c,e)\} \\ &= \max\{3, 11, 2\} \\ &= 11 \\ \text{diameter}(C_j) &= \max\{d(b,d)\} \\ &= 5 \end{aligned}$$

The cluster with the largest diameter is C_i. So we now split C_i.

We repeat the process by taking $C_l = \{a, c, e\}$. The remaining computations are left as an exercise to the reader.

13.12 Density-based clustering

In density-based clustering, clusters are defined as areas of higher density than the remainder of the data set. Objects in these sparse areas - that are required to separate clusters - are usually considered to be noise and border points. The most popular density based clustering method is DBSCAN (Density-Based Spatial Clustering of Applications with Noise).

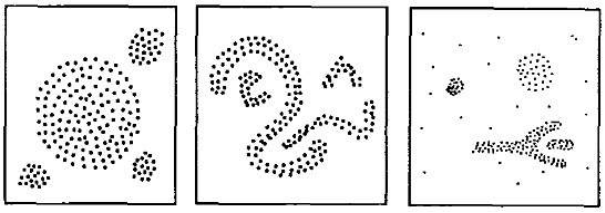


Figure 13.17: Clusters of points and noise points not belonging to any of those clusters

13.12.1 Density

We introduce some terminology and notations.

- Let ϵ (epsilon) be some constant distance. Let p be an arbitrary data point. The *ϵ-neighbourhood of p* is the set
$$N_\epsilon(p) = \{q : d(p,q) < \epsilon\}$$
- We choose some number m_0 to define points of "high density": We say that a point p is point of *high density* if $N_\epsilon(p)$ contains at least m_0 points.
- We define a point p as a *core point* if $N_\epsilon(p)$ has more than m_0 points.
- We define a point p as a *border point* if $N_\epsilon(p)$ has fewer than m_0 points, but is in the ϵ-neighbourhood of a core point.
- A point which is neither a core point nor a border point is called a *noise point*.

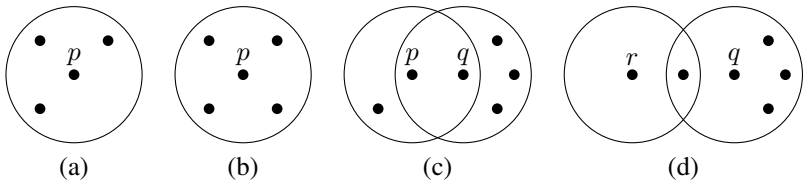


Figure 13.18: With $m_0 = 4$: (a) p a point of high density (b) p a core point (c) p a border point (d) r a noise point

- An object q is *directly density-reachable* from object p if p is a core object and q is in $N_\epsilon(p)$.
- An object q is *indirectly density-reachable* from an object p if there is a finite set of objects $p_1, \ldots, p_r$ such that p_1 is directly density-reachable form p, p_2 is directly density reachable from p_1, etc., q is directly density-reachable form p_r.

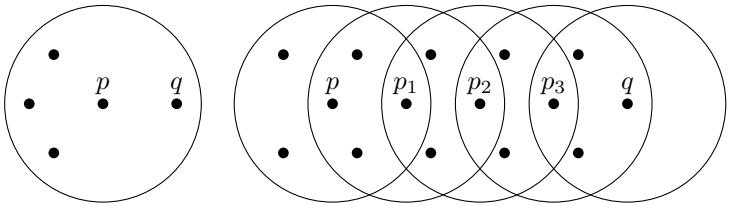


Figure 13.19: With $m_0 = 4$: (a) q is directly density-reachable from p (b) q is indirectly density-reachable from p

13.12.2 DBSCAN algorithm

Let $X = \{x_1, x_2, \ldots, x_n\}$ be the set of data points. DBSCAN requires two parameters: ϵ (eps) and the minimum number of points required to form a cluster (m_0).

Step 1. Start with an arbitrary starting point p that has not been visited.

Step 2. Extract the ϵ-neighborhood $N_\epsilon(p)$ of p.

Step 3. If the number of points in $N_\epsilon(p)$ is not greater than m_0 then the point p is labeled as noise (later this point can become the part of the cluster).

Step 4. If the number of points in $N_\epsilon(p)$ is greater than m_0 then the point p is a core point and is marked as visited. Select a new *cluster-id* and mark all objects in $N_\epsilon(p)$ with this cluster-id.

Step 5. If a point is found to be a part of the cluster then its ϵ-neighborhood is also the part of the cluster and the above procedure from step 2 is repeated for all ϵ-neighborhood points. This is repeated until all points in the cluster are determined.

Step 6. A new unvisited point is retrieved and processed, leading to the discovery of a further cluster or noise.

Step 7. This process continues until all points are marked as visited.

13.13 Sample questions

(a) Short answer questions

1. What is clustering?
2. Is clustering supervised learning? Why?
3. Explain some applications of the k-means algorithm.
4. Explain how clustering technique is used in image segmentation problem.
5. Explain how clustering technique used in data compression.
6. What is meant by the mixture of two normal distributions?
7. Explain hierarchical clustering.
8. What is a dendrogram? Give an example.
9. Is hierarchical clustering unsupervised learning? Why?
10. Describe the two methods for hierarchical clustering.

11. In a clustering problem, what does the measure of dissimilarity measure? Give some examples of measures of dissimilarity.

12. Explain the different types of linkages in clustering.

13. In the context of density-based clustering, define high density point, core point, border point and noise point.

14. What is agglomerative hierarchical clustering?

(b) Long answer questions

1. Apply k-means algorithm for given data with k = 3. Use $C_1(2)$, $C_2(16)$ and $C_3(38)$ as initial centers. Data:
$$2, 4, 6, 3, 31, 12, 15, 16, 38, 35, 14, 21, 3, 25, 30$$

2. Explain K-means algorithm and group the points (1, 0, 1), (1, 1, 0), (0, 0, 1) and (1, 1, 1) using K-means algorithm.

3. Applying the k-means algorithm, find two clusters in the following data.

x	185	170	168	179	182	188	180	180	183	180	180	177
y	72	56	60	68	72	77	71	70	84	88	67	76

4. Use k-means algorithm to find 2 clusters in the following data:

No.	1	2	3	4	5	6	7
x_1	1.0	1.5	3.0	5.0	3.5	4.5	3.5
x_2	1.0	2.0	4.0	7.0	5.0	5.0	4.5

5. Give a general outline of the expectation-maximization algorithm.

6. Describe EM algorithm for Gaussian mixtures.

7. Describe an algorithm for agglomerative hierarchical clustering.

8. Given the following distance matrix, construct the dendrogram using agglomerative clustering with single linkage, complete linkage and average linkage.

	A	B	C	D	E
A	0	1	2	2	3
B	1	0	2	4	3
C	2	2	0	1	5
D	2	4	1	0	3
E	3	3	5	3	0

9. Describe an algorithm for divisive hierarchical clustering.

10. For the data in Question 8, construct a dendrogram using DIANA algorithm.

11. Describe the DBSCAN algorithm for clustering.